

How Digital TV Manufacturers Can Tackle Chip Complexities

A unified modeling-verification-emulation-validation approach backed with domain expertise in audio/video and SoC architecture holds a strong promise



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Strike while the iron is hot! The global digital TV (DTV) market is red hot iron for original equipment manufacturers (OEMs) with about 300 million units expected to be sold by 2015. Even more radiant is the high-end DTV business commanding 20 per cent of the DTV market today.

The perfect digital TV today must provide connectivity, best-in-class video and graphics, seamless user experience, advanced features of auto-stereoscopic 3D or multi-/free-viewpoint TV, integration with social networks and wireless audio-visual stream, and a smart TV ecosystem. Innovation in the display technology and custom processing logic—traditionally

the forte of OEMs—alone can't create magic.

There is a need for new silicon platforms with multiple derivatives every 18 months, and in-depth understanding of complexities at the chip and software level to get the differentiating complex DTV system-on-chip (SoC) mould out faster. Simultaneously, there is mounting pressure on OEMs to become cost-effective with every product for different market segments.

Changing DTV landscape

Today's DTV doesn't comprise the olden-day chips where everything was done through a dedicated analogue circuitry with minimal central processing unit (CPU) processing and some digital audio/video accel-

eration. DTV SoCs have moved away from traditional DTV SoCs that had a single-core processor with a low-power graphics engine and capacity to handle 500 MIPS.

Block diagram of next-generation media SoC

Today, the complexity of the next-generation DTV media SoC has grown manifold. OEMs are looking at integrating most features into the DTV media SoC. This is resulting in design complexity shifting from the board design to the DTV media SoC.

The DTV media SoC has become the heart of DTV, just like an application processor powering a mobile phone. Alongside, more software is being encapsulated in the silicon today, making the chip behave more like a platform executing software along with serious hardware acceleration and post-processing assists. Software differentiation in this dynamic DTV landscape is fast becoming essential to preserve product differentiation. This calls for a need to scale the hardware accordingly.

These rising challenges are driving OEMs towards a next-generation complex DTV media SoC that would comprise a dual- or quad-core (ARM Cortex A-15) application processor, high-end graphics with multiple 3D and 2D cores, high-performance SoC configuration to support data throughput of $60 \text{ f/s} \times 2 = 120 \text{ f/s}$ for auto-stereoscopy, dedicated hardware blocks for video post-processing (like for scaling, filtering and enhancement), multi-format HD decoders and